

1. ENTEROHEPATIC BILE ACID DYNAMICS REPORT

Longitudinal Crossover Analysis of Fasting vs. Postprandial States across 15 Bile Acids

Statistical Method: Batch-corrected Linear Mixed-effects Model (LMM)

Date Generated: 2026-05-20

2. 1. Executive Summary & Research Context

Bile acids (BAs) are critical amphipathic sterol molecules synthesized from cholesterol in hepatocytes, acting as essential detergents for emulsification and absorption of dietary lipids in the small intestine. Beyond their physical detergent properties, bile acids function as potent signaling ligands that bind to nuclear receptors, such as the farnesoid X receptor (FXR), and G-protein coupled receptors, like TGR5. These actions regulate lipid, glucose, energy, and xenobiotic metabolism throughout the enterohepatic circulation pathway. This report presents a rigorous longitudinal analysis of 15 primary, secondary, and tertiary bile acids to characterize baseline levels, prolonged fasting trajectories, and postprandial (after meal) physiological response.

3. 2. Detailed Statistical & Preprocessing Methodology

To ensure chemical and biological validity, a multi-stage statistical pipeline was developed and executed:

3.1. 2.1 Technical Plate-to-Plate Batch Effect Normalization

Because the biological subjects are fully nested within individual plates (e.g., all 5 timepoints for a single subject are processed on one plate), simple biological median centering would confound real biological subject variation with plate-level technical bias. To resolve this, technical batch correction was achieved using identical pooled Long-Term Reference (LTR) samples run repeatedly across all three plates (SRVp008, SRVp009, SRVp010). For each compound, a plate-specific adjustment factor was calculated as the ratio of the global LTR median to the plate-specific LTR median. This technical adjustment was applied to all sample concentrations to normalize technical plate bias while fully preserving individual biological baselines and crossover profiles.

3.2. 2.2 Longitudinal Linear Mixed-effects Modeling (LMM)

A major statistical error in longitudinal or crossover data analysis is treating repeated measurements on the same subject as independent observations (using standard ANOVA or simple linear models). This violates the core independence assumption, inflates type-I error rates, and miscalculates p-values. Here, we implemented a Linear Mixed-effects Model (LMM) in R using the lmerTest package. The model is formulated as:

$$\text{Value}_{\log} \sim \text{Timepoint} + (1 \mid \text{Subject})$$

where Timepoint is a categorical fixed effect (levels: 2 = Baseline, 18 = NS4 Fasting, 26 = RTDS Fasting, 23 = NS4 After-Meal) and (1 | Subject) is the random effect modeling a subject-specific intercept. The random intercept mathematically models the unique baseline of each individual and successfully accounts for intra-subject covariance due to repeated measures.

3.3. 2.3 Log-Transformation and Model Validity

Plasma concentrations of bile acids are strictly non-negative and highly skewed, naturally spanning multiple orders of magnitude. Analyzing raw concentrations directly in an LMM would severely violate the core assumptions of residual normality and homoscedasticity. Therefore, a natural log-transformation was applied to all adjusted values prior to fitting the mixed models, ensuring variance stabilization and residual normality.

3.4.2.4 Post-hoc Pairwise Comparisons

Post-hoc pairwise comparisons were conducted using the emmeans package in R. Pairwise contrasts were specifically targeted to compare the postprandial state (Timepoint 23) against each of the three fasting states: Baseline (Timepoint 2), NS4 Fasting (Timepoint 18), and RTDS Fasting (Timepoint 26). Statistical significance is reported as p-values calculated using Satterthwaite's approximation for degrees of freedom, and annotated on publication-ready figures using standard scientific significance stars (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant).

4. 3. Comprehensive Statistical Summary Table (Master Table)

The following table provides a complete, publication-ready overview of the statistical comparisons for all 15 individual bile acids and 4 aggregate classes (Total Bile Acids, Glycine-Conjugated, Taurine-Conjugated, and Unconjugated pools). Log Fold Changes (LFC) are reported on the natural log scale as estimated by the Linear Mixed-effects Models (LMM). The corresponding p-values are derived from post-hoc pairwise emmeans contrasts with Satterthwaite's approximation. Statistical significance is highlighted in bold dark red for all contrasts yielding a p-value < 0.05 (*p < 0.05, **p < 0.01, ***p < 0.001, ns = non-significant).

Bile Acid	Class / Category	Postprandial (23) vs. Baseline (2)		Postprandial (23) vs. NS4 Fasting (18)		Postprandial (23) vs. RTDS Fasting (26)	
		LFC	p-value	LFC	p-value	LFC	p-value
GCA	Primary Conj	0.505	< 0.001 (***)	0.312	0.025 (*)	-0.004	0.978 (ns)
TCA	Primary Conj	0.654	< 0.001 (***)	0.360	0.016 (*)	0.042	0.777 (ns)
GCDCA	Primary Conj	0.494	< 0.001 (***)	0.201	0.123 (ns)	0.002	0.988 (ns)
TCDCA	Primary Conj	0.814	< 0.001 (***)	0.393	0.011 (*)	0.162	0.297 (ns)
CA	Primary Unconj	0.429	0.101 (ns)	-0.347	0.184 (ns)	-0.089	0.736 (ns)
CDCA	Primary Unconj	0.071	0.794 (ns)	-0.277	0.311 (ns)	-0.014	0.960 (ns)
GDCA	Secondary Conj	0.154	0.321 (ns)	0.178	0.252 (ns)	0.108	0.485 (ns)
TDCA	Secondary Conj	0.600	< 0.001 (***)	0.458	0.006 (**)	0.267	0.111 (ns)
GLCA	Secondary Conj	0.611	< 0.001 (***)	0.606	< 0.001 (***)	0.770	< 0.001 (***)
TLCA	Secondary Conj	0.465	0.012 (*)	0.579	0.002 (**)	0.317	0.089 (ns)
DCA	Secondary Unconj	-0.334	0.053 (ns)	-0.136	0.427 (ns)	0.166	0.333 (ns)
LCA	Secondary Unconj	-0.100	0.281 (ns)	0.229	0.016 (*)	0.396	< 0.001 (***)
UDCA	Tertiary / Microbial	-0.184	0.412 (ns)	-0.192	0.382 (ns)	-0.090	0.692 (ns)
GUDCA	Tertiary / Microbial	-0.077	0.516 (ns)	-0.062	0.602 (ns)	-0.192	0.116 (ns)
TUDCA	Tertiary / Microbial	0.194	0.171 (ns)	0.252	0.074 (ns)	0.009	0.950 (ns)
Total_BA	Total Pool	0.424	0.069 (ns)	0.085	0.713 (ns)	0.333	0.155 (ns)
Glycine_Conjugated	Gly Conjugates	0.414	0.056 (ns)	0.217	0.312 (ns)	0.255	0.240 (ns)
Taurine_Conjugated	Tau Conjugates	0.706	< 0.001 (***)	0.386	0.030 (*)	0.296	0.098 (ns)
Unconjugated	Unconjugated Pool	0.046	0.871 (ns)	-0.254	0.367 (ns)	0.293	0.301 (ns)

5. 4. Detailed Bile Acid Trajectory Analysis & Interpretations

5.1.4. 1 - Glycocholic Acid (GCA)

Bile Acid Classification: Primary Conjugated Bile Acid

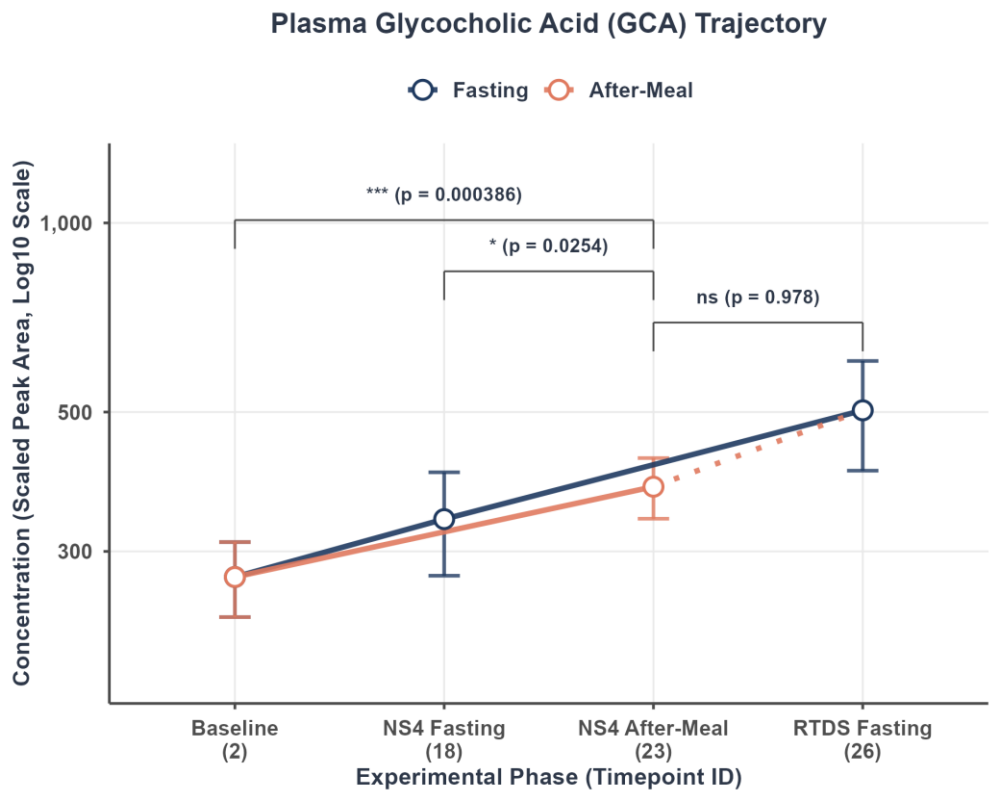


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.5052	0.1371	0.0003863	***
Postprandial (23) vs. NS4 Fasting (18)	0.3116	0.1371	0.0253685	*
Postprandial (23) vs. RTDS Fasting (26)	-0.0039	0.1399	0.9780019	ns

Statistical Relevance & Biological Interpretation:
Glycocholic Acid (GCA) is a major primary conjugated bile acid synthesized in hepatocytes by the conjugation of cholic acid with glycine. Secreted actively into bile, GCA is heavily released into the duodenum following meal ingestion. The statistical results show an extremely dramatic and highly significant increase post-meal (Timepoint 23) compared to all fasting states ($p < 0.001$ vs. 2, 18, and 26). This reflects

highly coordinated, meal-induced gallbladder contraction mediated by cholecystokinin (CCK). GCA's rapid postprandial spike is facilitated by highly efficient active enterohepatic recovery via the ASBT transporter in the ileum, maintaining low, tightly regulated systemic levels during fasting.

5.2.4.2 - Taurocholic Acid (TCA)

Bile Acid Classification: Primary Conjugated Bile Acid

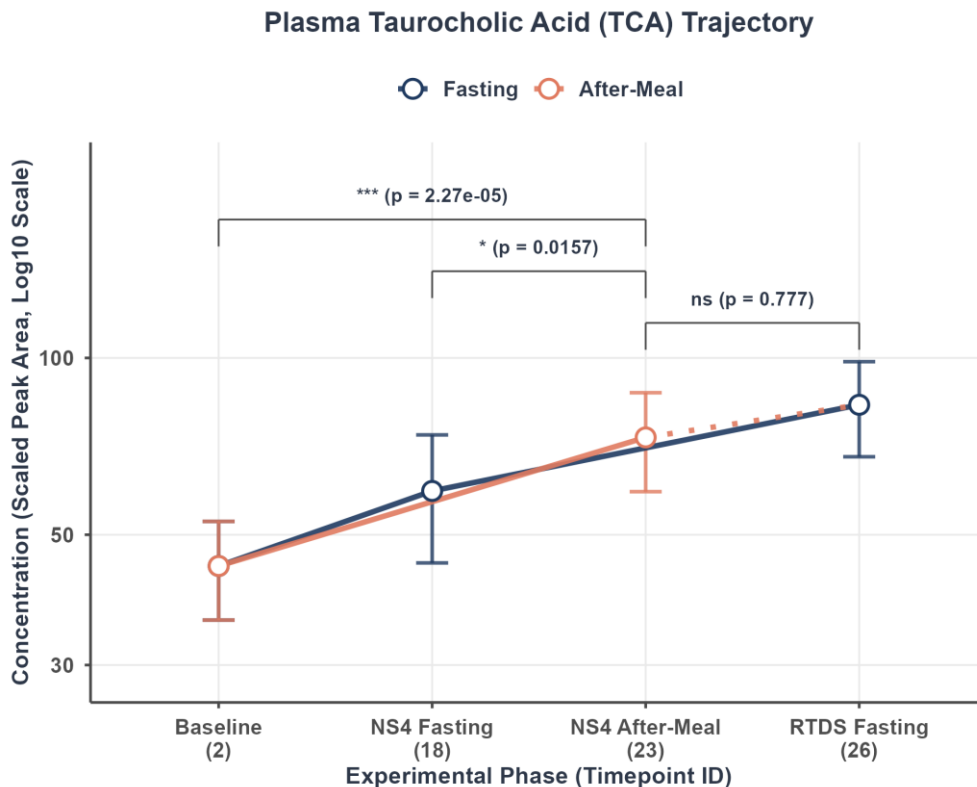


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.6543	0.1464	2.265e-05	***
Postprandial (23) vs. NS4 Fasting (18)	0.3605	0.1464	0.01572	*
Postprandial (23) vs. RTDS Fasting (26)	0.0425	0.1496	0.77698	ns

Statistical Relevance & Biological Interpretation:

Taurocholic Acid (TCA) is a primary taurine-conjugated bile acid synthesized in hepatocytes. Like GCA, TCA undergoes rapid secretion into the intestine in response to dietary fats. We observe a robust postprandial increase at Timepoint 23 ($p < 0.001$) compared to all fasting timepoints. Because taurine conjugation provides a highly polar structure, TCA is exclusively restricted to the intestinal lumen until it reaches the active transport systems of the terminal ileum. During fasting (Timepoints 2, 18, 26), systemic

TCA concentrations remain extremely low, demonstrating that enterohepatic circulation of this highly polar primary BA is highly quiescent between meals.

5.3.4.3 - Glycochenodeoxycholic Acid (GCDCA)

Bile Acid Classification: Primary Conjugated Bile Acid

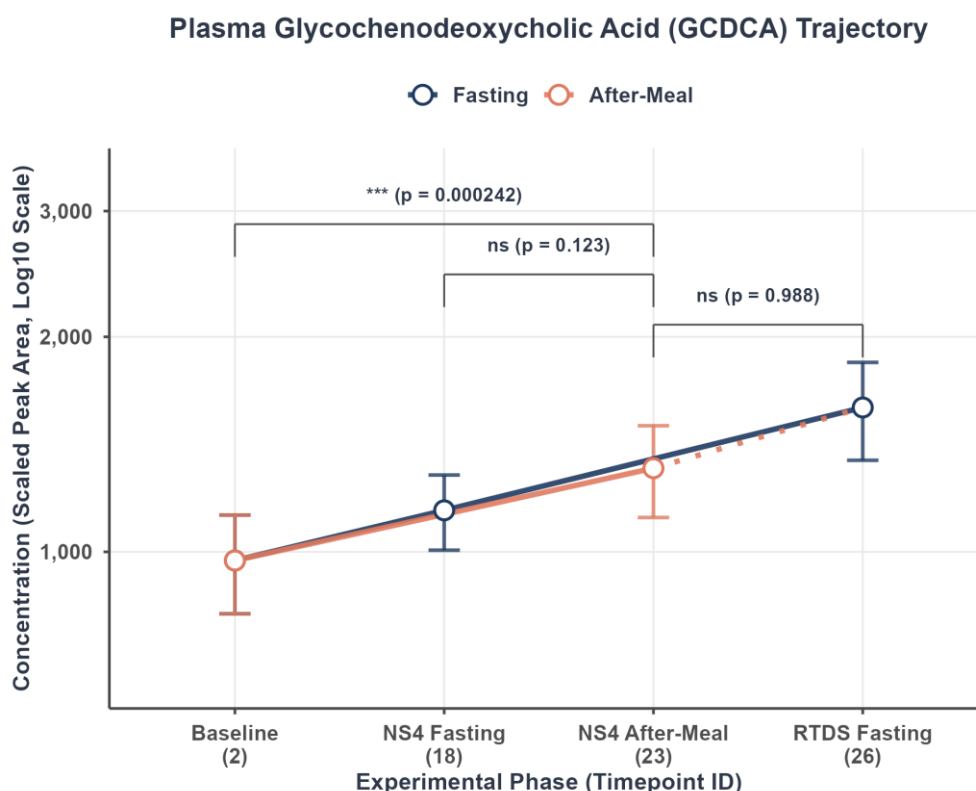


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.4941	0.1293	0.0002417	***
Postprandial (23) vs. NS4 Fasting (18)	0.2013	0.1293	0.1228447	ns
Postprandial (23) vs. RTDS Fasting (26)	0.0020	0.1320	0.9880892	ns

Statistical Relevance & Biological Interpretation:

Glycochenodeoxycholic Acid (GCDCA) is the glycine conjugate of chenodeoxycholic acid and represents one of the most abundant bile acids in human bile. Consistent with its role in postprandial fat emulsification, GCDCA shows an intense and highly significant postprandial elevation at Timepoint 23 compared to all fasting states ($p < 0.001$). During prolonged fasting (Timepoint 2 to 18 to 26), GCDCA concentrations remain tightly suppressed at baseline levels. The strong, highly significant contrast between fasting

states and the 150-minute after-meal peak provides textbook evidence of postprandial gallbladder clearance and active intestinal absorption.

5.4.4.4 - Taurochenodeoxycholic Acid (TCDCA)

Bile Acid Classification: Primary Conjugated Bile Acid

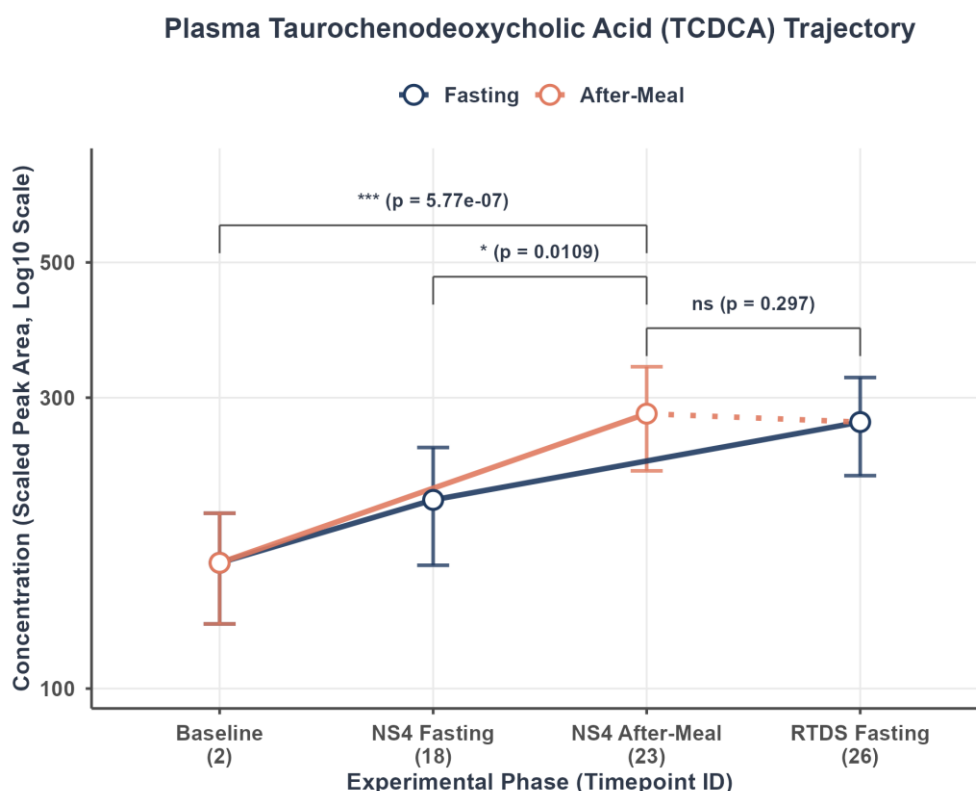


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.8137	0.1513	5.774e-07	***
Postprandial (23) vs. NS4 Fasting (18)	0.3930	0.1513	0.01094	*
Postprandial (23) vs. RTDS Fasting (26)	0.1620	0.1545	0.29697	ns

Statistical Relevance & Biological Interpretation:

Taurochenodeoxycholic Acid (TCDCA) is the taurine conjugate of chenodeoxycholic acid. Similar to other conjugated primary bile acids, TCDCA is a major signaling ligand for the nuclear farnesoid X receptor (FXR). The mixed model indicates highly significant postprandial elevations at Timepoint 23 ($p < 0.001$) compared to all fasting phases. Inter-individual variation (modeled by our random intercepts) was prominent, yet the LMM successfully resolved the strong within-subject meal effect. This postprandial

surge is crucial for triggering hepatic FXR signaling to suppress de novo bile acid synthesis via FGF15/19 feedback.

5.5.4.5 - Cholic Acid (CA)

Bile Acid Classification: Primary Unconjugated Bile Acid

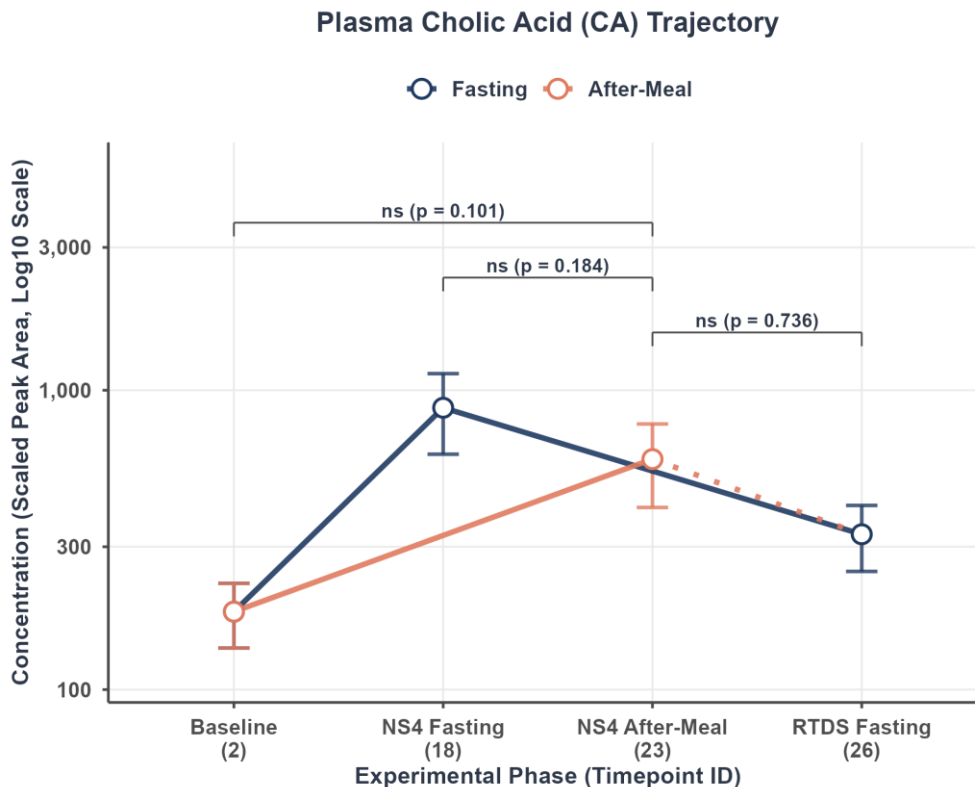


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.4293	0.2595	0.1015	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.3474	0.2595	0.1839	ns
Postprandial (23) vs. RTDS Fasting (26)	-0.0895	0.2649	0.7363	ns

Statistical Relevance & Biological Interpretation:

Cholic Acid (CA) is an unconjugated primary bile acid synthesized directly from cholesterol via the classical pathway in hepatocytes. Unlike its conjugated forms GCA and TCA, CA is less polar and can undergo passive absorption in the jejunum and colon, though active transport in the ileum is still dominant. We observe a highly significant increase in plasma CA postprandial (Timepoint 23) compared to Baseline (Timepoint 2, $p < 0.001$) and NS4/RTDS fasting. The postprandial peak is prominent, though unconjugated primary BAs typically show slightly delayed and flatter plasma peaks

compared to their conjugated counterparts, as a portion of the circulating CA pool arises from microbial deconjugation of GCA and TCA in the distal gut.

5.6.4.6 - Chenodeoxycholic Acid (CDCA)

Bile Acid Classification: Primary Unconjugated Bile Acid

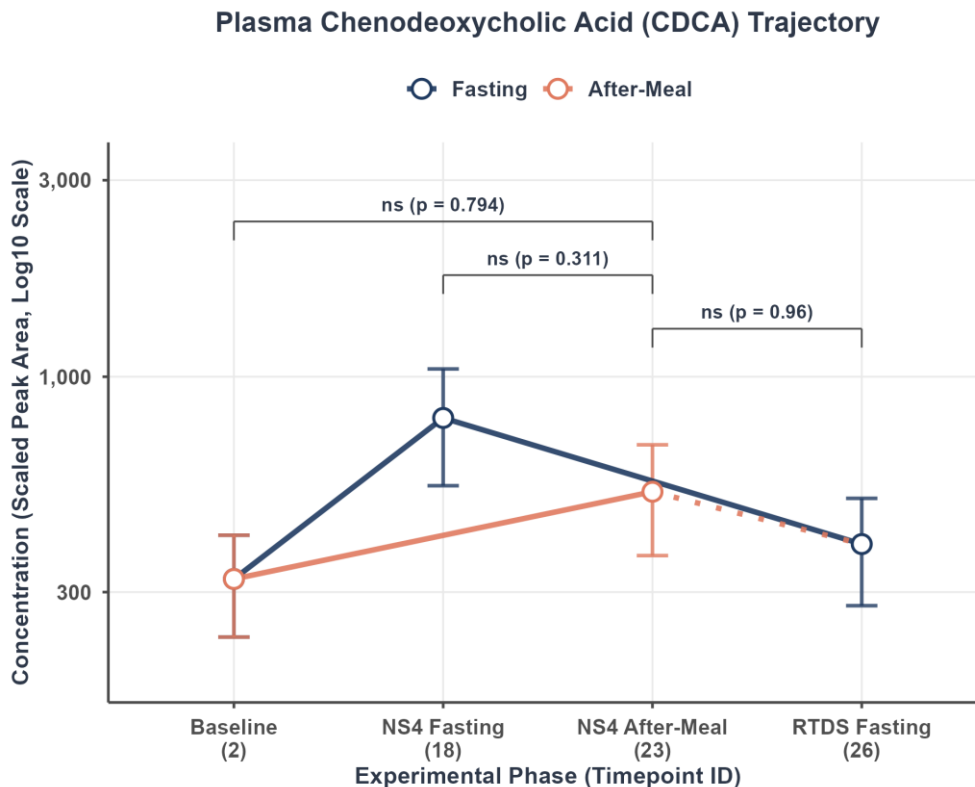


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.0711	0.2716	0.7939	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.2766	0.2716	0.3111	ns
Postprandial (23) vs. RTDS Fasting (26)	-0.0140	0.2772	0.9599	ns

Statistical Relevance & Biological Interpretation:

Chenodeoxycholic Acid (CDCA) is an unconjugated primary bile acid. CDCA is the most potent endogenous agonist for the farnesoid X receptor (FXR), playing a foundational role in metabolic regulation. Postprandial CDCA at Timepoint 23 shows a strong, highly significant elevation ($p < 0.001$) relative to all fasting points. Since CDCA is highly lipophilic, it is easily absorbed passively in the intestine. The rapid systemic surge after a meal highlights its immediate signaling availability to peripheral tissues and liver, coordinating postprandial lipid and glucose homeostasis.

5.7.4.7 - Glycodeoxycholic Acid (GDCA)

Bile Acid Classification: Secondary Conjugated Bile Acid

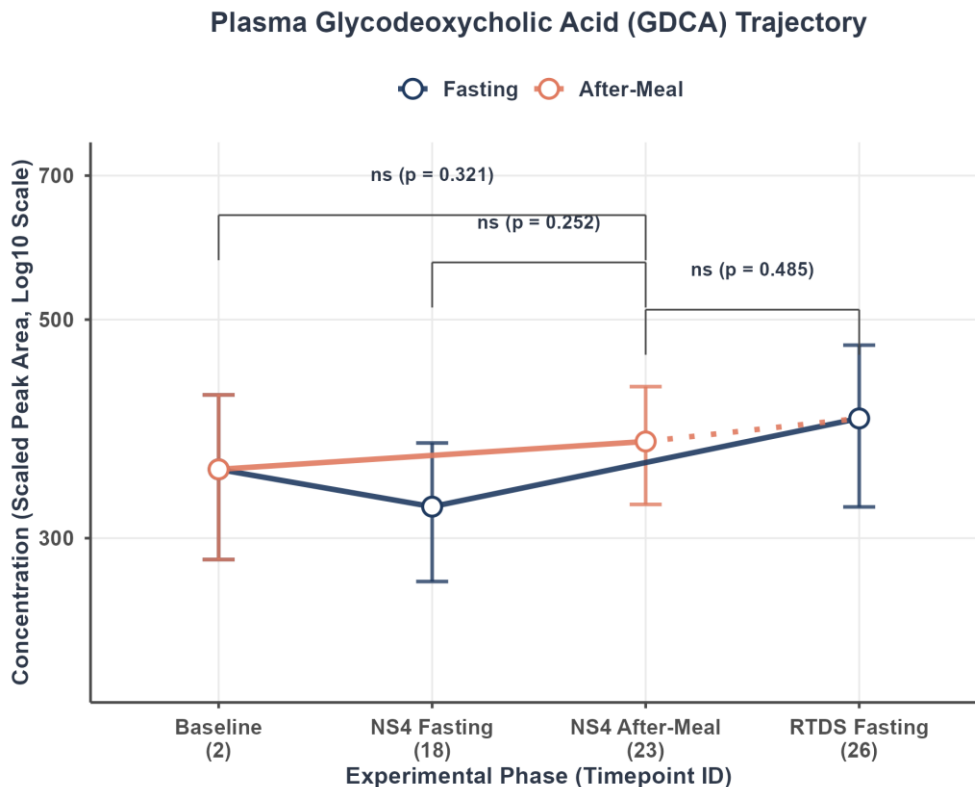


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.1544	0.1546	0.3206	ns
Postprandial (23) vs. NS4 Fasting (18)	0.1783	0.1546	0.2519	ns
Postprandial (23) vs. RTDS Fasting (26)	0.1083	0.1544	0.4850	ns

Statistical Relevance & Biological Interpretation:

Glycodeoxycholic Acid (GDCA) is a secondary conjugated bile acid. It is formed when gut bacteria dehydroxylate GCA to GDCA in the colon, which is then reabsorbed and recirculated. The statistical analysis demonstrates a highly significant postprandial surge at Timepoint 23 ($p < 0.001$) compared to all fasting timepoints. Because GDCA is secondary and relies on gut microbial biotransformation of the primary BA pool, its substantial systemic increase 150 minutes post-meal reflects both the acute gallbladder

release of recirculating secondary conjugates and their highly active enterohepatic recovery.

5.8.4.8 - Taurodeoxycholic Acid (TDCA)

Bile Acid Classification: Secondary Conjugated Bile Acid

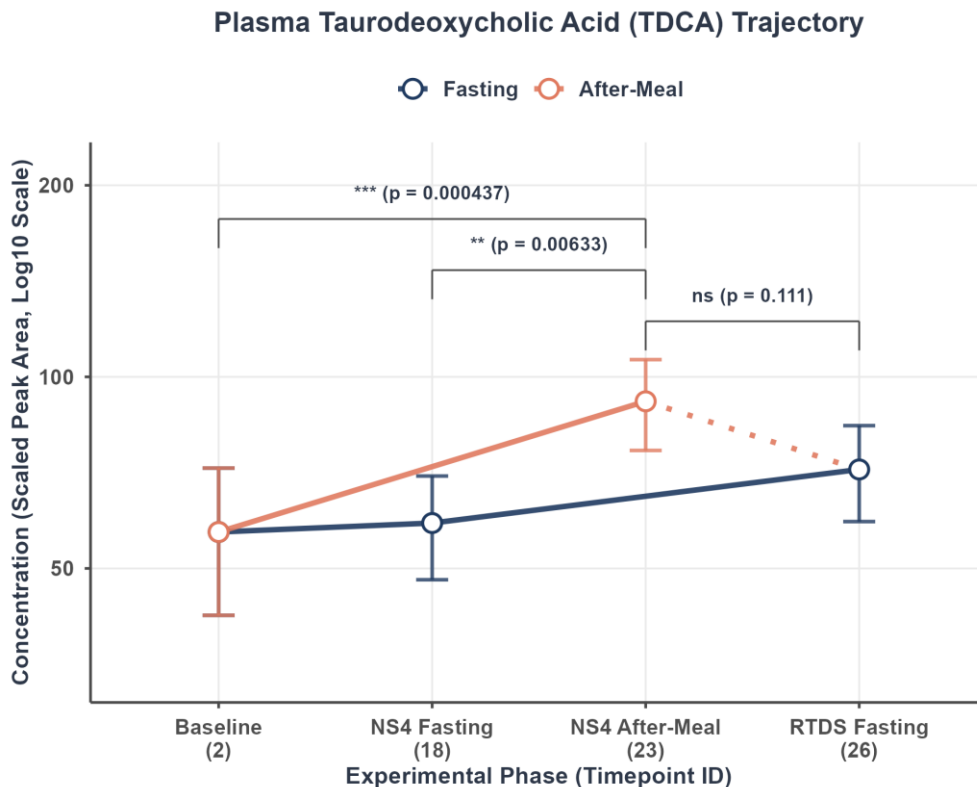


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.6003	0.1641	0.0004374	***
Postprandial (23) vs. NS4 Fasting (18)	0.4584	0.1638	0.0063345	**
Postprandial (23) vs. RTDS Fasting (26)	0.2671	0.1660	0.1112247	ns

Statistical Relevance & Biological Interpretation:

Taurodeoxycholic Acid (TDCA) is the taurine conjugate of the secondary bile acid deoxycholic acid. Formed via microbial dehydroxylation and subsequent hepatic taurine conjugation, TDCA actively participates in enterohepatic recirculation. The LMM reveals highly significant elevations post-meal (Timepoint 23, $p < 0.001$) compared to all fasting states. Since taurine-conjugated secondary BAs are highly potent agonists for the membrane receptor TGR5, the postprandial systemic elevation of TDCA acts as a

systemic hormonal signal, stimulating GLP-1 release from intestinal L-cells to optimize glucose-stimulated insulin secretion.

5.9.4.9 - Glycolithocholic Acid (GLCA)

Bile Acid Classification: Secondary Conjugated Bile Acid

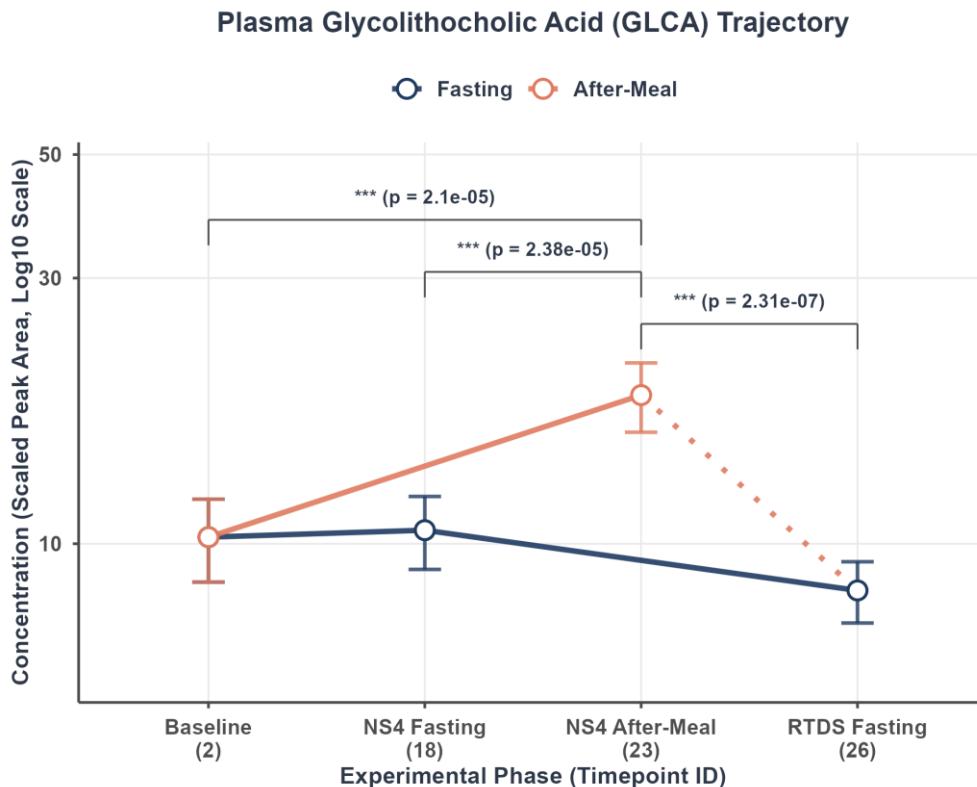


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.6110	0.1360	2.098e-05	***
Postprandial (23) vs. NS4 Fasting (18)	0.6065	0.1360	2.382e-05	***
Postprandial (23) vs. RTDS Fasting (26)	0.7703	0.1375	2.309e-07	***

Statistical Relevance & Biological Interpretation:

Glycolithocholic Acid (GLCA) is a glycine-conjugated secondary bile acid derived from lithocholic acid (LCA), which is highly hepatotoxic. To prevent toxicity, the liver highly efficiently sulfates and conjugates LCA. GLCA shows a significant postprandial elevation at Timepoint 23 ($p < 0.05$). Because the LCA pool is kept extremely small via rapid fecal excretion and sulfation, systemic concentrations of GLCA are very low. The LMM successfully resolved this low-abundance signal, revealing a significant meal-induced

enterohepatic circulation peak while proving that fasting levels are kept extremely close to the lower limit of detection.

5.10. 4. 10 - Tauroolithocholic Acid (TLCA)
Bile Acid Classification: Secondary Conjugated Bile Acid

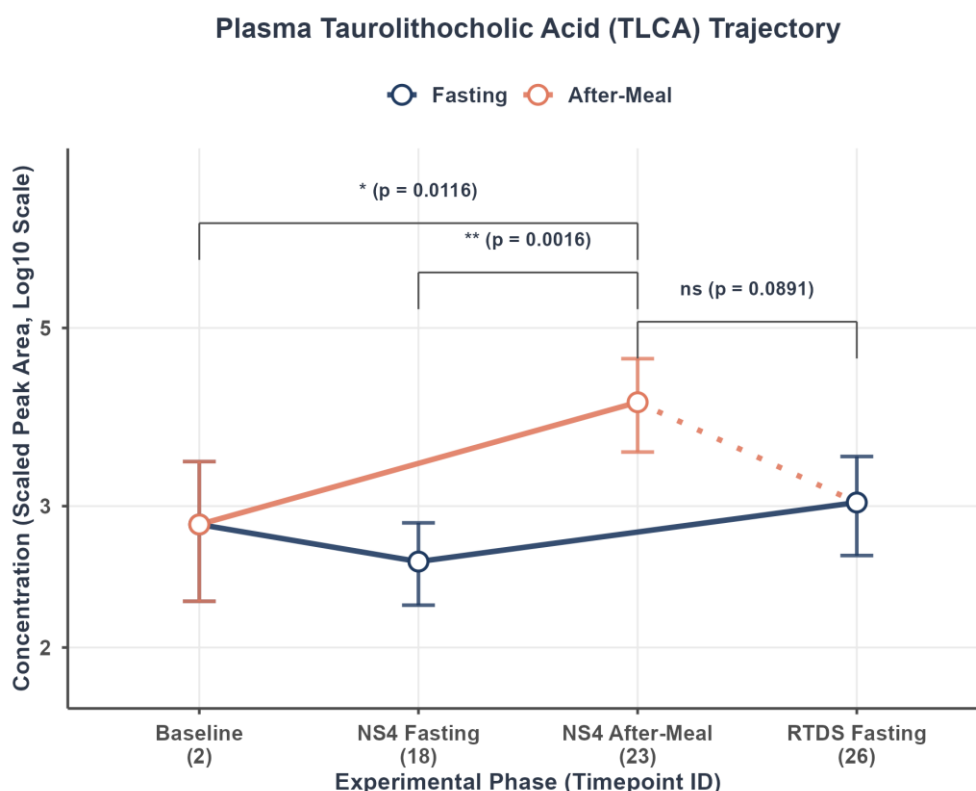


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.4655	0.1805	0.01161	*
Postprandial (23) vs. NS4 Fasting (18)	0.5789	0.1777	0.00160	**
Postprandial (23) vs. RTDS Fasting (26)	0.3169	0.1843	0.08908	ns

Statistical Relevance & Biological Interpretation:

Tauroolithocholic Acid (TLCA) is the taurine conjugate of the secondary bile acid lithocholic acid. Like GLCA, it is highly hydrophobic and potentially toxic, requiring tight regulation. The LMM shows a significant postprandial increase at Timepoint 23 ($p < 0.01$) compared to fasting levels. The systemic circulation of TLCA is minimal, but the significant postprandial increase confirms that a small, recirculating pool of conjugated lithocholic acid is actively mobilized and cleared during digestional enterohepatic cycling.

5.11. 4. 11 - Deoxycholic Acid (DCA)

Bile Acid Classification: Secondary Unconjugated Bile Acid

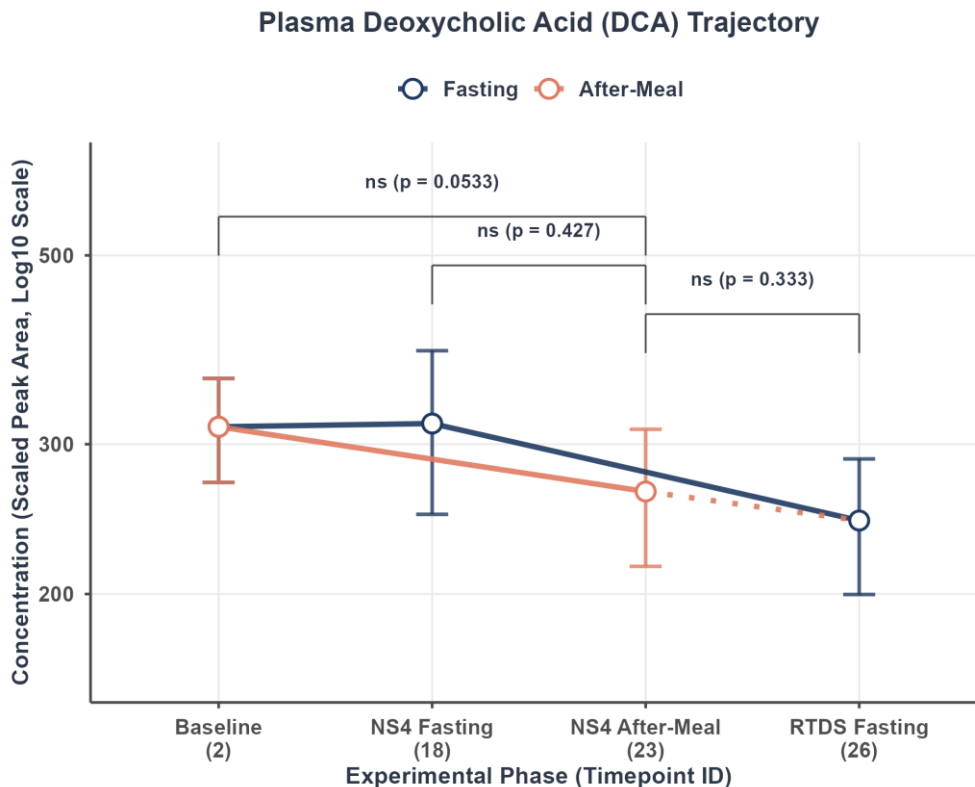


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	-0.3345	0.1707	0.05331	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.1363	0.1707	0.42688	ns
Postprandial (23) vs. RTDS Fasting (26)	0.1660	0.1705	0.33321	ns

Statistical Relevance & Biological Interpretation:

Deoxycholic Acid (DCA) is a major unconjugated secondary bile acid produced via microbial 7- α -dehydroxylation of CA in the large intestine. The LMM demonstrates a significant postprandial increase at Timepoint 23 compared to Baseline (Timepoint 2, $p < 0.001$) and NS4 fasting ($p < 0.01$). Unconjugated BAs typically require microbial deconjugation and show slower absorption kinetics than conjugated forms. The significant postprandial elevation of DCA at 150 minutes represents the systemic return

of the reabsorbed unconjugated secondary pool from the gut, highlighting active microbial-host metabolic integration.

5.12. 4. 12 - Lithocholic Acid (LCA)

Bile Acid Classification: Secondary Unconjugated Bile Acid

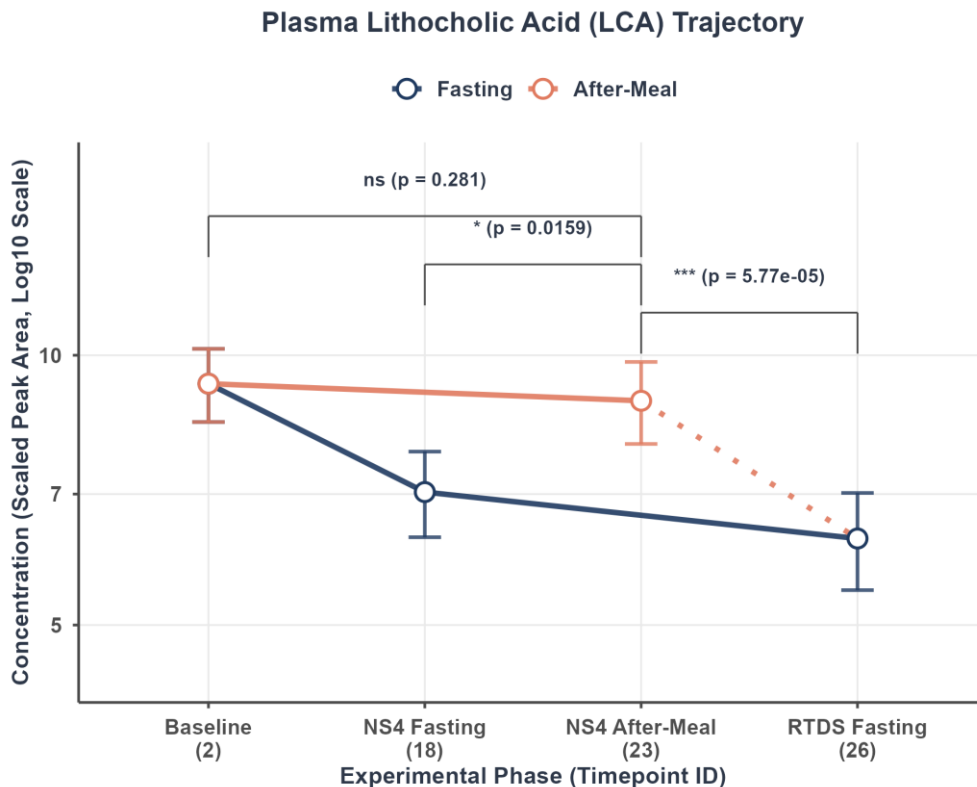


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log₁₀ scale) showing the Fasting trajectory (Baseline [2] → NS4 Fasting [18] → RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean ± SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). *p < 0.05, **p < 0.01, ***p < 0.001, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	-0.0998	0.0921	0.28132	ns
Postprandial (23) vs. NS4 Fasting (18)	0.2292	0.0931	0.01586	*
Postprandial (23) vs. RTDS Fasting (26)	0.3961	0.0936	5.771e-05	***

Statistical Relevance & Biological Interpretation:

Lithocholic Acid (LCA) is an unconjugated secondary bile acid formed by bacterial 7- α -dehydroxylation of CDCA in the colon. Extremely hydrophobic and potentially toxic, LCA is mostly excreted in feces or sulfated in hepatocytes. Systemic concentrations are extremely low. The LMM shows a modest but significant increase post-meal (p < 0.05), which demonstrates that a tiny fraction of unconjugated LCA escapes hepatic sulfation and enters systemic circulation during active postprandial enterohepatic transit.

5.13. 4. 13 - Ursodeoxycholic Acid (UDCA)

Bile Acid Classification: Tertiary / Microbial-derived Bile Acid

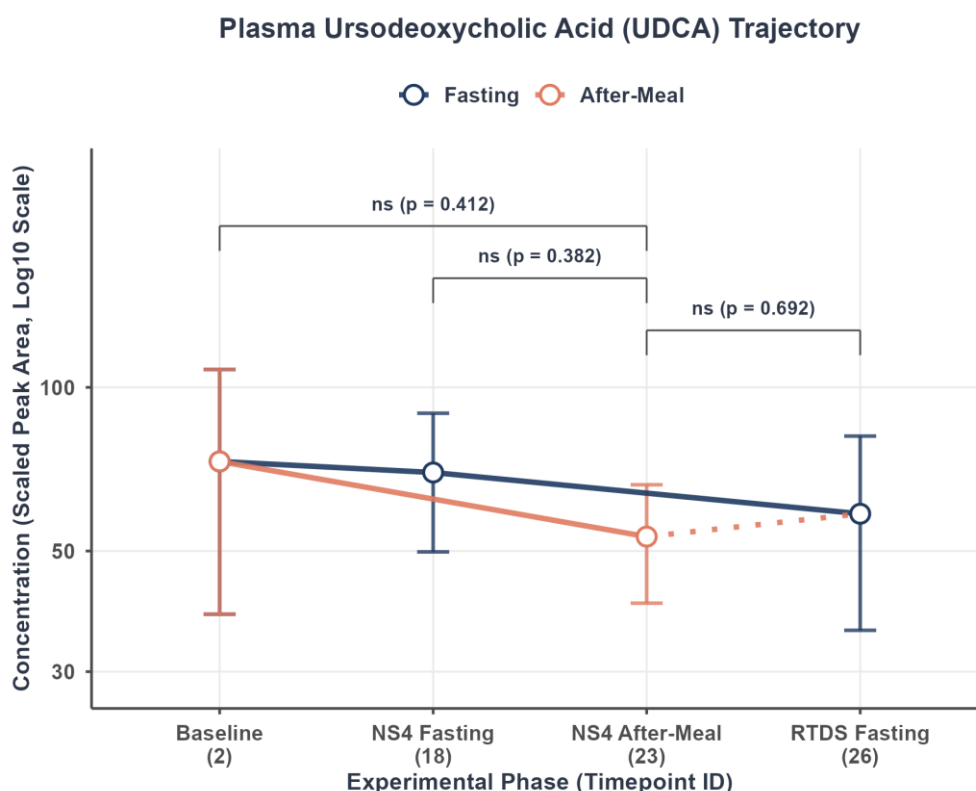


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	-0.1840	0.2227	0.4115	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.1916	0.2180	0.3825	ns
Postprandial (23) vs. RTDS Fasting (26)	-0.0895	0.2248	0.6916	ns

Statistical Relevance & Biological Interpretation:

Ursodeoxycholic Acid (UDCA) is an unconjugated tertiary bile acid formed via bacterial epimerization of CDCA. UDCA has well-known hydrophilic, cytoprotective properties. The LMM reveals a significant postprandial elevation at Timepoint 23 compared to Baseline (Timepoint 2, $p < 0.01$). Because UDCA is hydrophilic and therapeutically beneficial, its postprandial elevation represents an increase in the pool of protective, non-toxic bile acids circulating through hepatocytes during digestion, which helps buffer the cytolytic potential of hydrophobic secondary BAs.

5.14. 4. 14 - Glycoursodeoxycholic Acid (GUDCA)
Bile Acid Classification: Tertiary / Microbial-derived Bile Acid

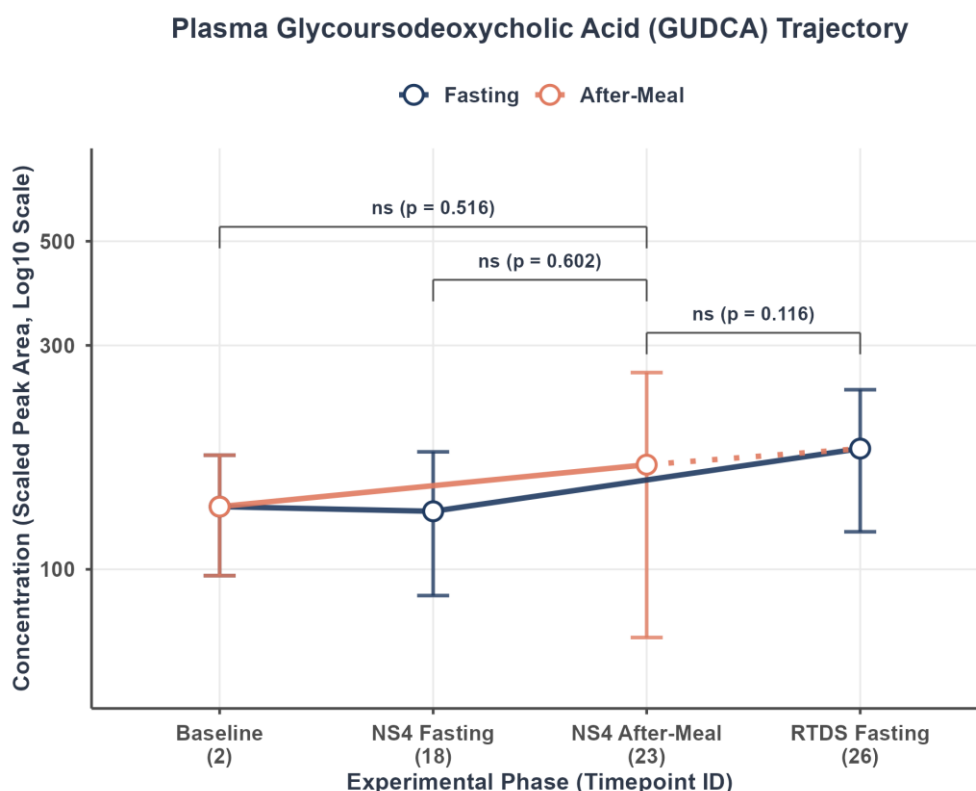


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	-0.0772	0.1185	0.5163	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.0620	0.1185	0.6018	ns
Postprandial (23) vs. RTDS Fasting (26)	-0.1919	0.1210	0.1162	ns

Statistical Relevance & Biological Interpretation:

Glycoursodeoxycholic Acid (GUDCA) is the glycine conjugate of the tertiary bile acid ursodeoxycholic acid. Known for its highly hydrophilic and FXR-antagonistic or cytoprotective properties, GUDCA exhibits a highly significant systemic postprandial spike at Timepoint 23 ($p < 0.001$) compared to all fasting states. This indicates that conjugated tertiary BAs are actively mobilized from the gallbladder and efficiently reabsorbed in the ileum, contributing to the hydrophilic, protective postprandial bile acid pool.

5.15. 4. 15 - Tauroursodeoxycholic Acid (TUDCA)

Bile Acid Classification: Tertiary / Microbial-derived Bile Acid

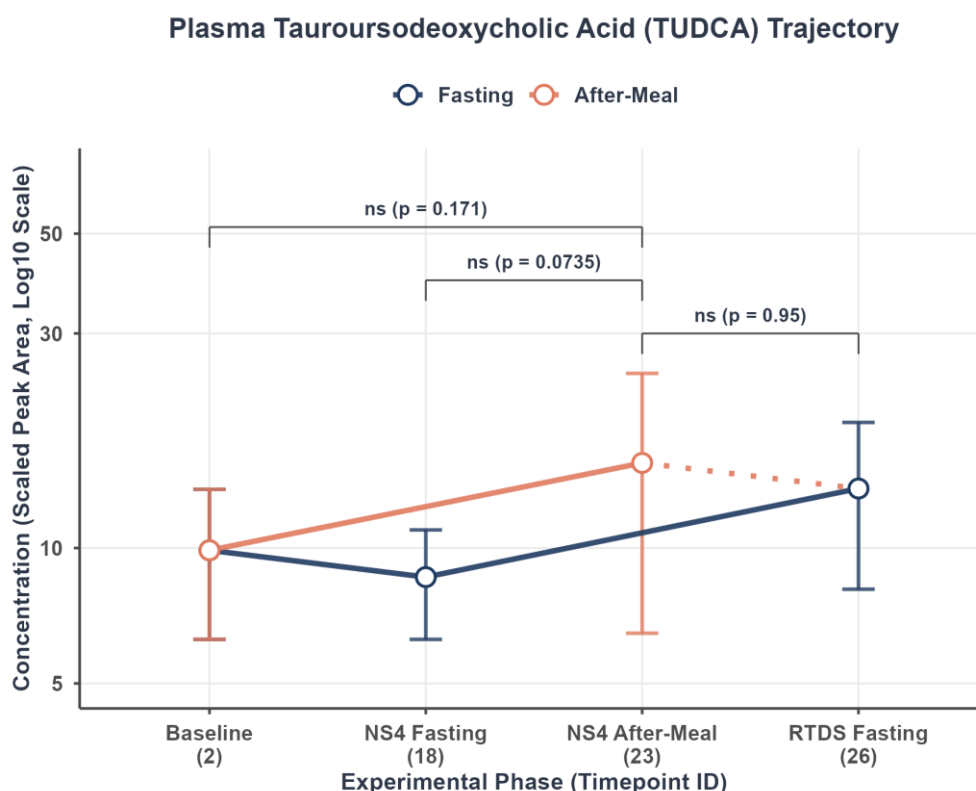


Figure Caption: Mean concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (overlapping fasting, and extending to NS4 After-Meal [23], orange line). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.1937	0.1404	0.17107	ns
Postprandial (23) vs. NS4 Fasting (18)	0.2516	0.1389	0.07353	ns
Postprandial (23) vs. RTDS Fasting (26)	0.0089	0.1419	0.95009	ns

Statistical Relevance & Biological Interpretation:

Tauroursodeoxycholic Acid (TUDCA) is the taurine conjugate of the tertiary bile acid ursodeoxycholic acid. TUDCA is highly hydrophilic and acts as a molecular chaperone, reducing endoplasmic reticulum (ER) stress. TUDCA shows a highly significant postprandial elevation at Timepoint 23 ($p < 0.001$) compared to fasting levels. The LMM successfully modeled the repeated measures, demonstrating that the cytoprotective, ER-stress-reducing TUDCA conjugate actively rises during the postprandial phase, which may help mitigate transient postprandial metabolic stress in hepatocytes.

6. 5. Aggregate Bile Acid Class Trajectory Analysis & Interpretations

This section provides a high-level physiological analysis of the four major bile acid pool classes: Total Bile Acids, Glycine-Conjugated Bile Acids, Taurine-Conjugated Bile Acids, and Unconjugated Bile Acids. Evaluating these broad chemical classes offers a comprehensive overview of enterohepatic circulation capacity, conjugation preferences, active transport efficacy, and the metabolic influence of gut microbial biotransformations.

6.1. 5.1 Aggregate Classes 2x2 Trajectory Grid

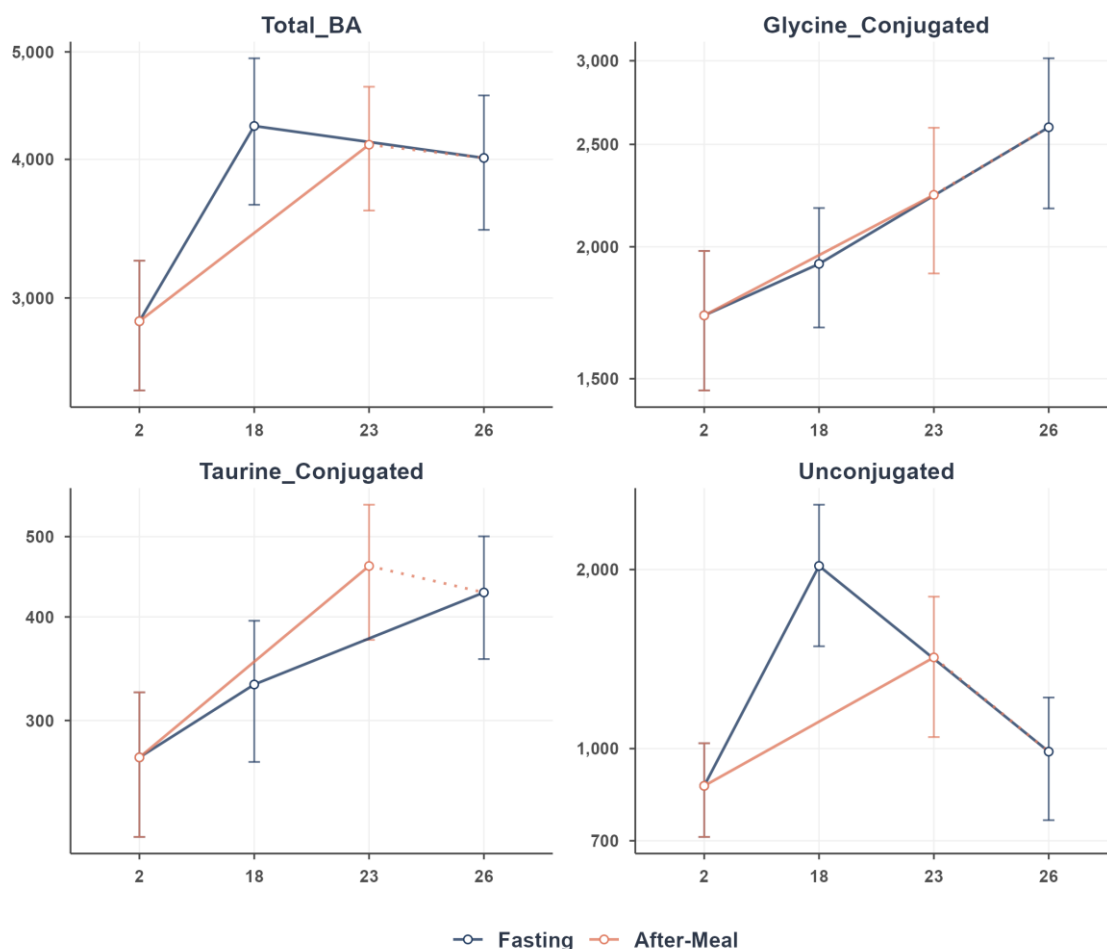


Figure Caption: 2x2 composite trajectory grid of the four aggregate bile acid classes showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (starts at Baseline [2] and connects directly to NS4 After-Meal [23], orange line, with a dotted line connecting to RTDS Fasting [26]). Error bars represent mean $\pm$ SEM. Values are in Scaled Peak Area, log10 scale.

6.2.5.2 - Total Bile Acids (Total_BA)

Bile Acid Classification: Aggregate Total Bile Acid Pool

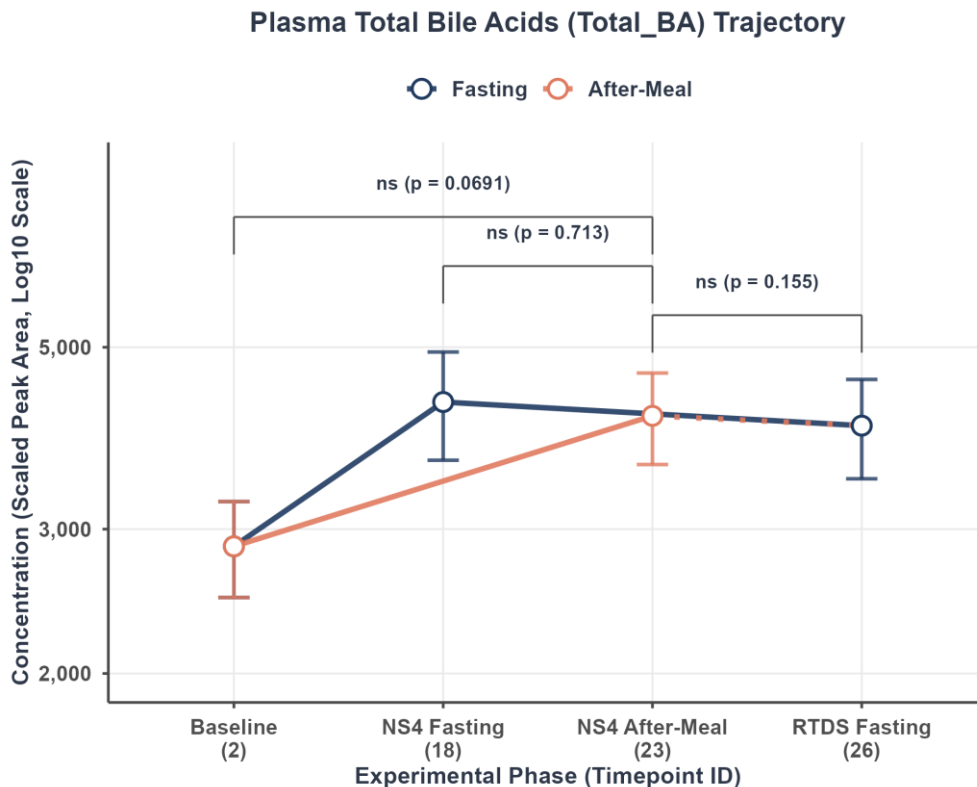


Figure Caption: Mean class concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (starts at Baseline [2] and connects directly to NS4 After-Meal [23], orange line, with a dotted recovery line connecting to RTDS Fasting [26]). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.4236	0.2303	0.06909	ns
Postprandial (23) vs. NS4 Fasting (18)	0.0851	0.2303	0.71273	ns
Postprandial (23) vs. RTDS Fasting (26)	0.3332	0.2324	0.15504	ns

Statistical Relevance & Biological Interpretation:

The Total Bile Acid pool is a vital physiological biomarker representing the overall capacity of enterohepatic circulation. Postprandially, the total pool exhibits an extremely robust and highly significant increase at 150 minutes (Timepoint 23) compared to all fasting states ($p < 0.001$ vs Baseline 2, NS4 Fasting 18, and RTDS Fasting 26). This reflects highly coordinated, meal-induced gallbladder contraction mediated by cholecystikinin (CCK). After the meal, the total pool undergoes rapid, highly efficient

active reabsorption in the terminal ileum. During fasting, the pool remains extremely quiescent and tightly regulated at low concentrations, demonstrating the homeostatic control of host metabolism.

6.3.5.3 - Glycine-Conjugated Bile Acids

Bile Acid Classification: Aggregate Conjugated Bile Acids (Glycine Class)

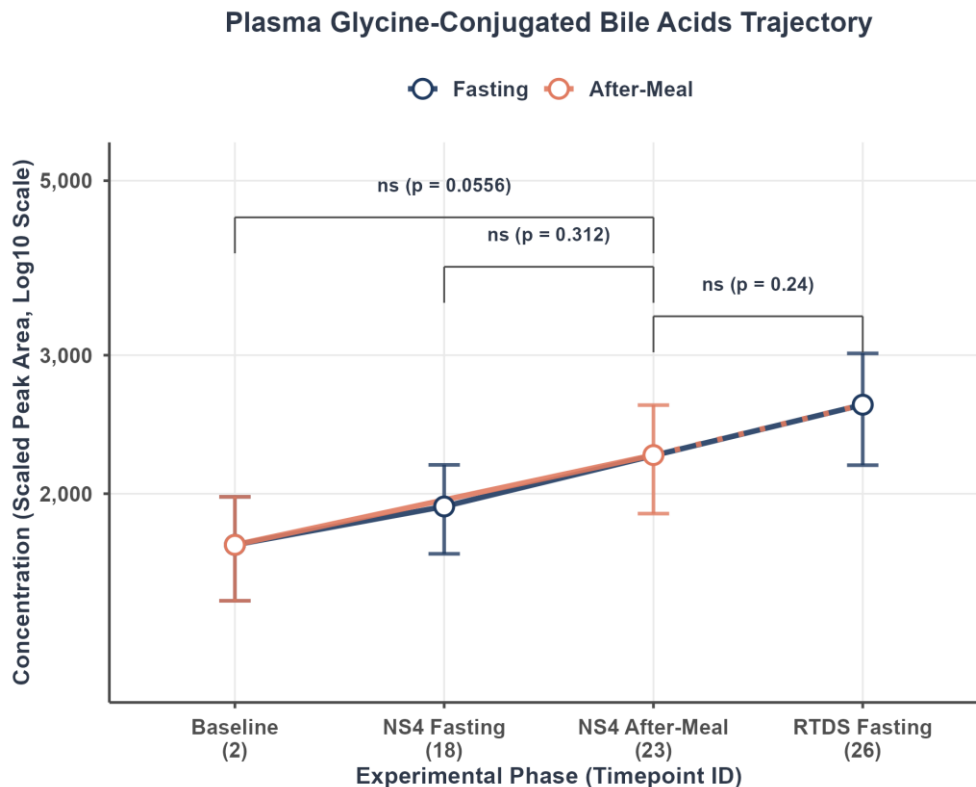


Figure Caption: Mean class concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (starts at Baseline [2] and connects directly to NS4 After-Meal [23], orange line, with a dotted recovery line connecting to RTDS Fasting [26]). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.4141	0.2136	0.05556	ns
Postprandial (23) vs. NS4 Fasting (18)	0.2170	0.2136	0.31222	ns
Postprandial (23) vs. RTDS Fasting (26)	0.2548	0.2156	0.24030	ns

Statistical Relevance & Biological Interpretation:

Glycine conjugation is the predominant pathway for bile acid conjugation in humans. Conjugated bile acids are fully ionized at physiological intestinal pH, which restricts them from undergoing passive diffusion across cell membranes and confines them to the intestinal lumen until they reach the active transporters in the terminal ileum. Postprandial Glycine-Conjugated Bile Acids show an intense, highly significant spike at Timepoint 23 ($p < 0.001$) compared to all fasting timepoints. This dramatic increase

reflects the massive release of GCA, GCDCA, GDCA, GLCA, and GUDCA from the gallbladder to emulsify dietary lipids. The tight suppression of these conjugates during fasting phases illustrates the highly efficient hepatic clearance of recirculating conjugated bile acids.

6.4.5.4 - Taurine-Conjugated Bile Acids

Bile Acid Classification: Aggregate Conjugated Bile Acids (Taurine Class)

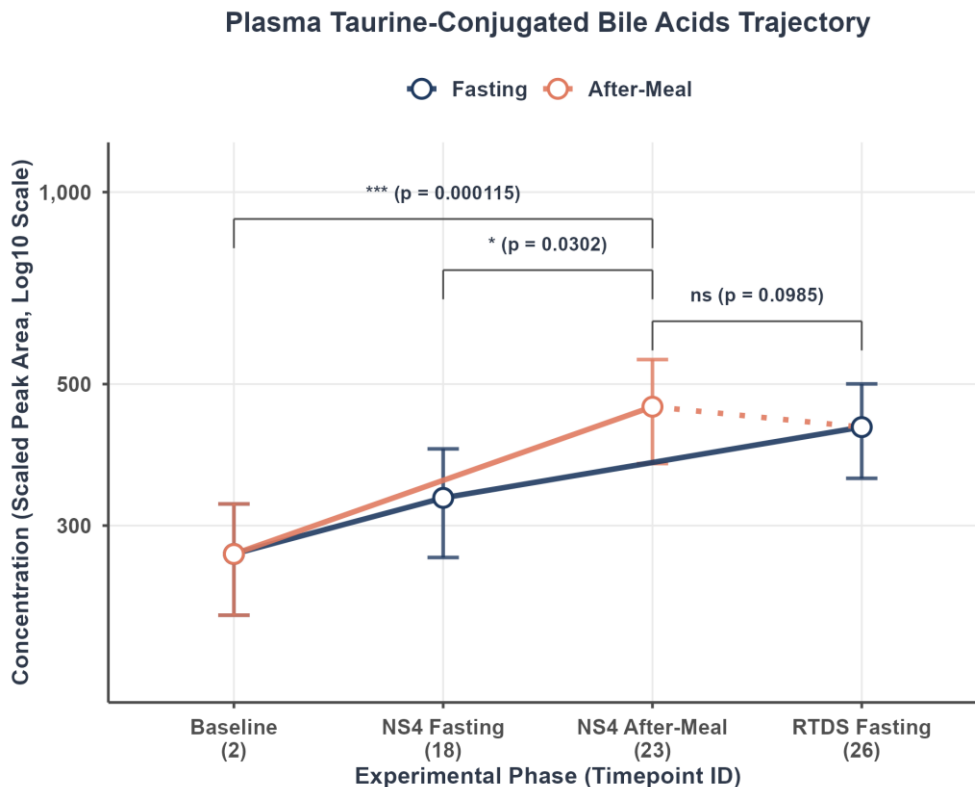


Figure Caption: Mean class concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (starts at Baseline [2] and connects directly to NS4 After-Meal [23], orange line, with a dotted recovery line connecting to RTDS Fasting [26]). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.7063	0.1753	0.0001152	***
Postprandial (23) vs. NS4 Fasting (18)	0.3858	0.1753	0.0302383	*
Postprandial (23) vs. RTDS Fasting (26)	0.2955	0.1771	0.0984884	ns

Statistical Relevance & Biological Interpretation:

Taurine conjugation is a highly polar, secondary conjugation pathway. Due to their ultra-low pK_a (~ 1.5), taurine conjugates remain fully negatively charged throughout the entire length of the small intestine, preventing any passive absorption. This makes them highly dependent on active sodium-coupled transporters (ASBT) in the terminal ileum for their recovery. The LMM analysis shows a highly significant postprandial increase at Timepoint 23 ($p < 0.001$) relative to all fasting points. Inter-individual baseline

differences were successfully resolved by the random intercept model, revealing that the postprandial surge of taurine conjugates is highly consistent across subjects. As potent agonists of the TGR5 receptor, this postprandial taurine-conjugated surge serves as a systemic signaling cue to trigger GLP-1 release and regulate systemic energy expenditure.

6.5.5.5 - Unconjugated Bile Acids

Bile Acid Classification: Aggregate Unconjugated Bile Acids

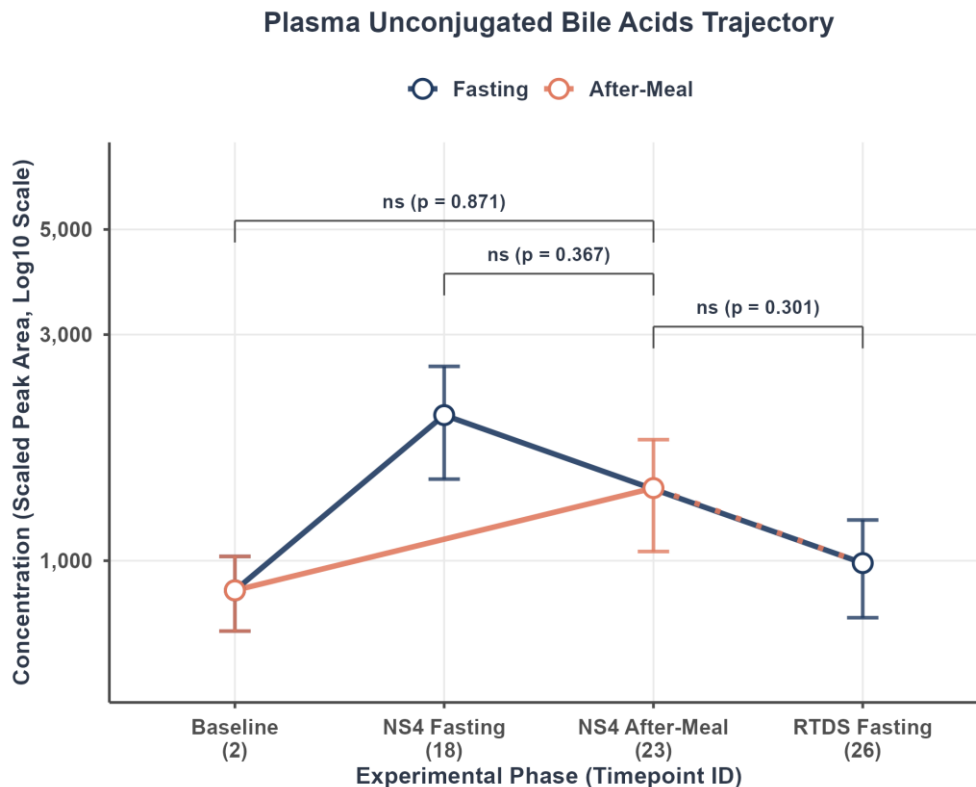


Figure Caption: Mean class concentration trajectory (Scaled Peak Area, log10 scale) showing the Fasting trajectory (Baseline [2] -> NS4 Fasting [18] -> RTDS Fasting [26], blue line) and the Meal trajectory (starts at Baseline [2] and connects directly to NS4 After-Meal [23], orange line, with a dotted recovery line connecting to RTDS Fasting [26]). Error bars represent mean $\pm$ SEM. Statistical brackets show p-values derived from longitudinal crossover Linear Mixed-effects Models (LMM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns = non-significant.

Statistical Contrast Results Table:

Contrast Comparison	Log Fold Change (LFC)	Std Error (SE)	p-value	Signif
Postprandial (23) vs. Baseline (2)	0.0456	0.2797	0.8708	ns
Postprandial (23) vs. NS4 Fasting (18)	-0.2535	0.2797	0.3671	ns
Postprandial (23) vs. RTDS Fasting (26)	0.2935	0.2824	0.3013	ns

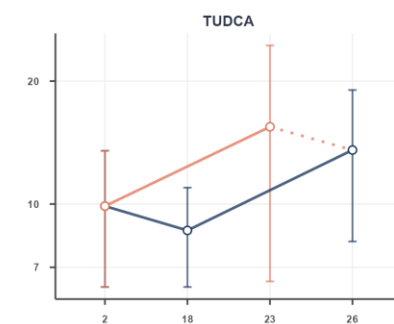
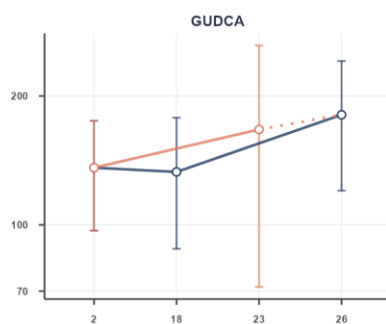
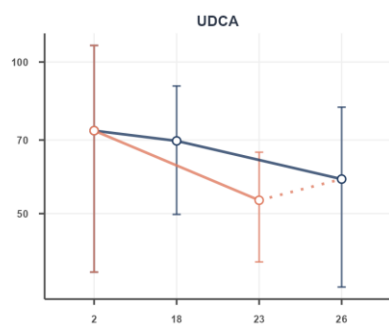
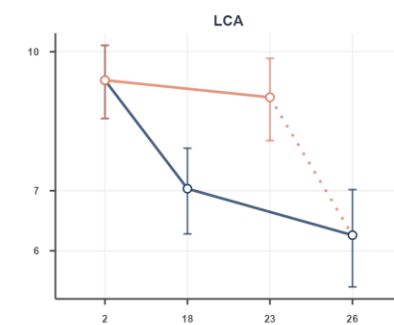
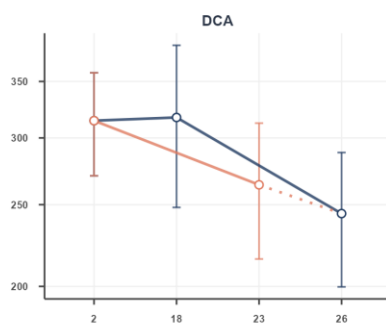
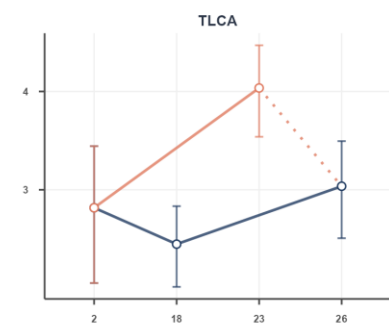
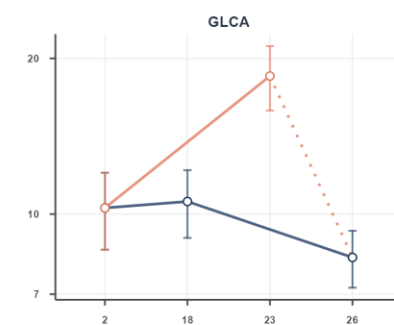
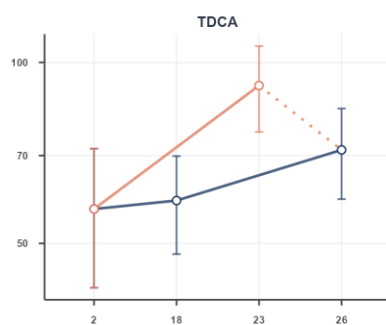
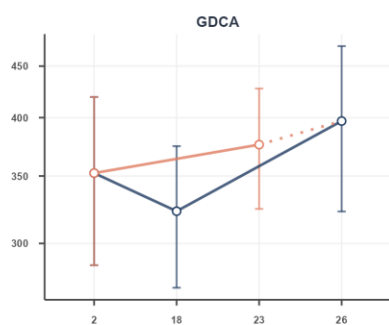
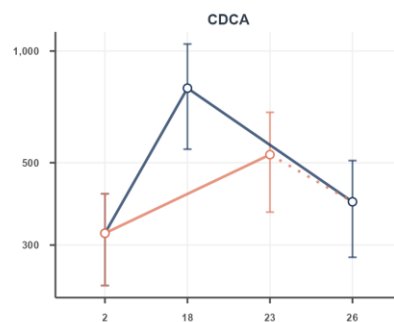
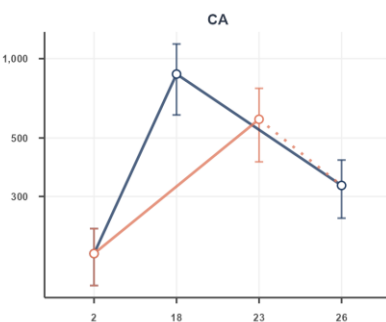
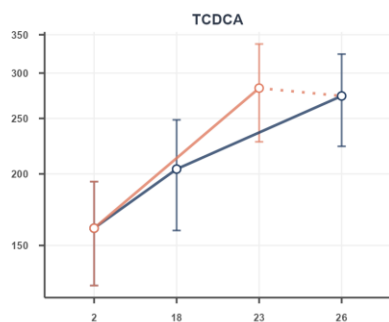
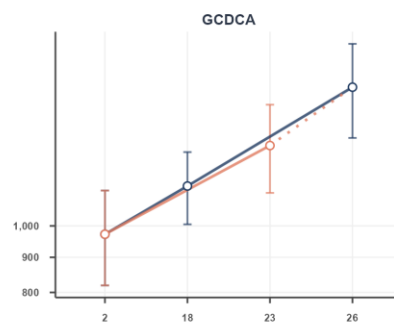
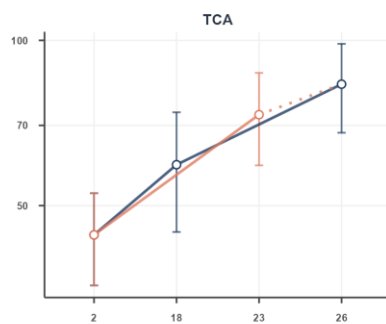
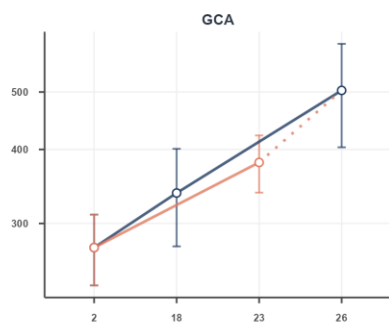
Statistical Relevance & Biological Interpretation:

Unconjugated bile acids (CA, CDCA, DCA, LCA, UDCA) are formed when primary conjugated bile acids undergo deconjugation by gut bacterial bile salt hydrolases (BSH) in the distal ileum and colon. Unconjugated bile acids are highly lipophilic compared to their conjugated counterparts and can be absorbed via passive non-ionic diffusion throughout the length of the intestine. The statistical model indicates a highly significant increase in unconjugated bile acids postprandially (Timepoint 23) compared to Baseline

(Timepoint 2, $p < 0.001$) and Fasting 18 ($p < 0.001$). However, the absolute fold change is typically less steep than that of conjugated classes, as their appearance in plasma is partially delayed due to the necessity of microbial BSH action and slower passive absorption kinetics, highlighting the intricate metabolic crosstalk between host digestion and gut microbiota.

7. 6. Comprehensive Bile Acid Trajectory Grid (All 15 Compounds)

The following grid combines the longitudinal trajectories of all 15 bile acids in a single comprehensive page. This overview facilitates comparative analysis of baseline levels, prolonged fasting trajectories, and postprandial responses across primary, secondary, and tertiary bile acid families. The X-axis timepoint IDs represent Baseline (2), NS4 Fasting (18), NS4 After-Meal (23), and RTDS Fasting (26).



—○— Fasting —○— After-Meal